**9.**

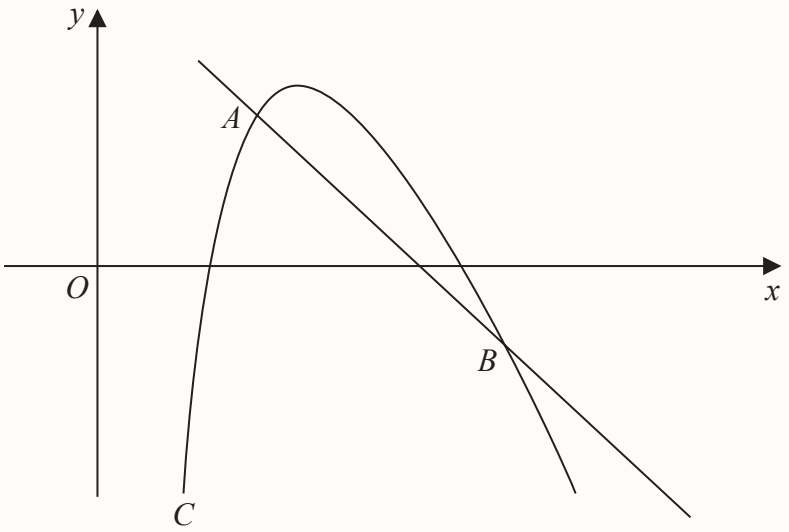



**Figure 3**

A sketch of part of the curve $C$ with equation

$$y = 20 - 4x - \frac{18}{x}, \qquad x > 0$$

is shown in Figure 3.

Point $A$ lies on $C$ and has $x$ coordinate equal to 2

(a) Show that the equation of the normal to $C$ at $A$ is $y = -2x + 7$.

**(6)**

The normal to $C$ at $A$ meets $C$ again at the point $B$, as shown in Figure 3.

(b) Use algebra to find the coordinates of $B$.

**(5)**